

SHETH L.U.J. AND SIR M.V. COLLEGE

Implementation Of Research Using R

CODE:-

```
install.packages("caret")  
install.packages("e1071")  
install.packages("ggplot2")  
install.packages("pROC")  
install.packages("dplyr")  
install.packages("smotefamily")
```

```
library(caret)  
library(e1071)  
library(ggplot2)  
library(pROC)  
library(dplyr)  
library(smotefamily)
```

```
data <- read.csv("heart.csv")
```

```
data$target <- as.factor(data$target)
```

```
set.seed(123)
```

```
trainIndex <- createDataPartition(data$target, p = 0.7, list = FALSE)  
train <- data[trainIndex, ]  
test <- data[-trainIndex, ]
```

NAME:- CHETAN MANDAVKAR

ROLL NO. S093

SUBJECT:- Data Analysis With SAS / SPSS / R

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```
model_raw <- svm(target ~ ., data = train, probability = TRUE)

pred_raw <- predict(model_raw, test, probability = TRUE)

cm_raw <- confusionMatrix(pred_raw, test$target)

accuracy_raw <- as.numeric(cm_raw$overall["Accuracy"])

cat("\nAccuracy WITHOUT Preprocessing:", round(accuracy_raw,3), "\n")
```

```
preProc_impute <- preProcess(train, method = "medianImpute")
train_imputed <- predict(preProc_impute, train)
test_imputed <- predict(preProc_impute, test)

preProc_scale <- preProcess(train_imputed, method = c("center", "scale"))
train_scaled <- predict(preProc_scale, train_imputed)
test_scaled <- predict(preProc_scale, test_imputed)
```

```
train_x <- train_scaled[, colnames(train_scaled) != "target"]
train_y <- train_scaled$target
```

```
train_y_num <- as.numeric(as.character(train_y))
```

```
smote_output <- SMOTE(train_x, train_y_num, K = 5)
```

```
train_balanced <- smote_output$data
```

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```
colnames(train_balanced)[ncol(train_balanced)] <- "target"
```

```
train_balanced$target <- as.factor(train_balanced$target)
```

```
model_processed <- svm(target ~ .,  
                        data = train_balanced,  
                        probability = TRUE)
```

```
pred_processed <- predict(model_processed,  
                          test_scaled,  
                          probability = TRUE)
```

```
cm_processed <- confusionMatrix(pred_processed, test_scaled$target)
```

```
accuracy_processed <- as.numeric(cm_processed$overall["Accuracy"])
```

```
cat("Accuracy WITH Preprocessing:", round(accuracy_processed,3), "\n")
```

```
accuracy_df <- data.frame(  
  Model = c("Raw Data", "Preprocessed Data"),  
  Accuracy = c(accuracy_raw, accuracy_processed)  
)
```

```
ggplot(accuracy_df, aes(x = Model, y = Accuracy, fill = Model)) +  
  geom_bar(stat = "identity", width = 0.6) +  
  coord_cartesian(ylim = c(0,1)) +  
  geom_text(aes(label = round(Accuracy,3)),  
            vjust = -0.5, size = 5) +
```

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```
ggtitle("Impact of Data Preprocessing on Prediction Accuracy") +  
ylab("Accuracy") +  
xlab("Model") +  
theme_minimal() +  
theme(legend.position = "none")
```

```
prob_raw <- attr(pred_raw, "probabilities")[,2]
```

```
roc_raw <- roc(response = test$target,  
  predictor = prob_raw,  
  levels = c("0","1"),  
  direction = "<")
```

```
prob_processed <- attr(pred_processed, "probabilities")[,2]
```

```
roc_processed <- roc(response = test_scaled$target,  
  predictor = prob_processed,  
  levels = c("0","1"),  
  direction = "<")
```

```
plot(roc_raw, col="red", main="ROC Curve Comparison")  
plot(roc_processed, col="blue", add=TRUE)
```

```
legend("bottomright",  
  legend=c("Raw Data","Preprocessed Data"),  
  col=c("red","blue"),  
  lwd=2)
```

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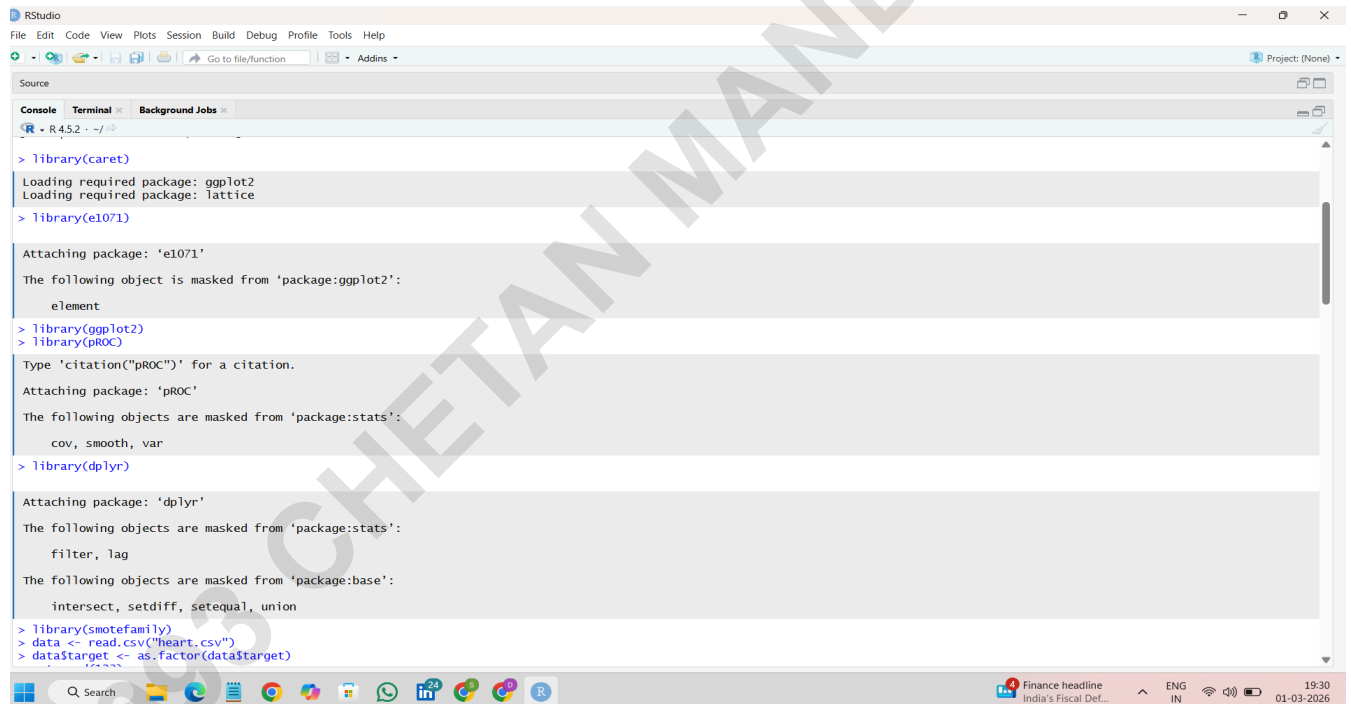
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```
cat("\nAUC Raw:", round(auc(roc_raw),3), "\n")
cat("AUC Processed:", round(auc(roc_processed),3), "\n")
```

```
cat("\nConfusion Matrix - Raw Data\n")
print(cm_raw)
```

```
cat("\nConfusion Matrix - Preprocessed Data\n")
print(cm_processed)
```

OUTPUT:-



```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins Project: (None)

Source
Console Terminal Background Jobs

> library(caret)
Loading required package: ggplot2
Loading required package: lattice
> library(e1071)
Attaching package: 'e1071'
The following object is masked from 'package:ggplot2':
  element
> library(ggplot2)
> library(pROC)
Type 'citation("pROC")' for a citation.
Attaching package: 'pROC'
The following objects are masked from 'package:stats':
  cov, smooth, var
> library(dplyr)
Attaching package: 'dplyr'
The following objects are masked from 'package:stats':
  filter, lag
The following objects are masked from 'package:base':
  intersect, setdiff, setequal, union
> library(smotefamily)
> data <- read.csv("heart.csv")
> data$target <- as.factor(data$target)
```

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```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins Project: (None)

Source
Console Terminal Background Jobs

R - R4.5.2 - ~/
> library(smoteFamily)
> data <- read.csv("heart.csv")
> data$target <- as.factor(data$target)
> set.seed(123)
> trainIndex <- createDataPartition(data$target, p = 0.7, list = FALSE)
> train <- data[trainIndex, ]
> test <- data[-trainIndex, ]
> model_raw <- svm(target ~ ., data = train, probability = TRUE)
> pred_raw <- predict(model_raw, test, probability = TRUE)
> cm_raw <- confusionMatrix(pred_raw, test$target)
> accuracy_raw <- as.numeric(cm_raw$overall["Accuracy"])
> cat("\nAccuracy WITHOUT Preprocessing:", round(accuracy_raw,3), "\n")

Accuracy WITHOUT Preprocessing: 0.822
> preProc_impute <- preProcess(train, method = "medianImpute")
> train_imputed <- predict(preProc_impute, train)
> test_imputed <- predict(preProc_impute, test)
> preProc_scale <- preProcess(train_imputed, method = c("center", "scale"))
> train_scaled <- predict(preProc_scale, train_imputed)
> test_scaled <- predict(preProc_scale, test_imputed)
> train_x <- train_scaled[, colnames(train_scaled) != "target"]
> train_y <- train_scaled$target
> train_y_num <- as.numeric(as.character(train_y))
> smote_output <- SMOTE(train_x, train_y_num, K = 5)

There were 50 or more warnings (use warnings() to see the first 50)

> train_balanced <- smote_output$data
> colnames(train_balanced)[ncol(train_balanced)] <- "target"
> train_balanced$target <- as.factor(train_balanced$target)
> model_processed <- svm(target ~ .,
+ data = train_balanced,
+ probability = TRUE)
> pred_processed <- predict(model_processed,
+ test_scaled,
+ probability = TRUE)
> cm_processed <- confusionMatrix(pred_processed, test_scaled$target)
> accuracy_processed <- as.numeric(cm_processed$overall["Accuracy"])
> cat("Accuracy WITH Preprocessing:", round(accuracy_processed,3), "\n")

Accuracy WITH Preprocessing: 0.822
```

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins Project: (None)

Source
Console Terminal Background Jobs

R - R4.5.2 - ~/
> cat("Accuracy WITH Preprocessing:", round(accuracy_processed,3), "\n")
Accuracy WITH Preprocessing: 0.822
> accuracy_df <- data.frame(
+ Model = c("Raw Data", "Preprocessed Data"),
+ Accuracy = c(accuracy_raw, accuracy_processed)
+ )
> ggplot(accuracy_df, aes(x = Model, y = Accuracy, fill = Model)) +
+ geom_bar(stat = "identity", width = 0.6) +
+ coord_cartesian(ylim = c(0,1)) +
+ geom_text(aes(label = round(Accuracy,3)),
+ vjust = -0.5, size = 5) +
+ ggtitle("Impact of Data Preprocessing on Prediction Accuracy") +
+ ylab("Accuracy") +
+ xlab("Model") +
+ theme_minimal() +
+ theme(legend.position = "none")
> prob_raw <- attr(pred_raw, "probabilities")[,2]
> roc_raw <- roc(response = test$target,
+ predictor = prob_raw,
+ levels = c("0", "1"),
+ direction = "<")
> prob_processed <- attr(pred_processed, "probabilities")[,2]
> roc_processed <- roc(response = test_scaled$target,
+ predictor = prob_processed,
+ levels = c("0", "1"),
+ direction = "<")
> plot(roc_raw, col="red", main="ROC Curve Comparison")
> plot(roc_processed, col="blue", add=TRUE)
> legend("bottomright",
+ legend=c("Raw Data", "Preprocessed Data"),
+ col=c("red", "blue"),
+ lwd=2)
> cat("\nAUC Raw:", round(auc(roc_raw),3), "\n")

AUC Raw: 0.102
> cat("AUC Processed:", round(auc(roc_processed),3), "\n")
AUC Processed: 0.898
> cat("\nConfusion Matrix - Raw Data\n")

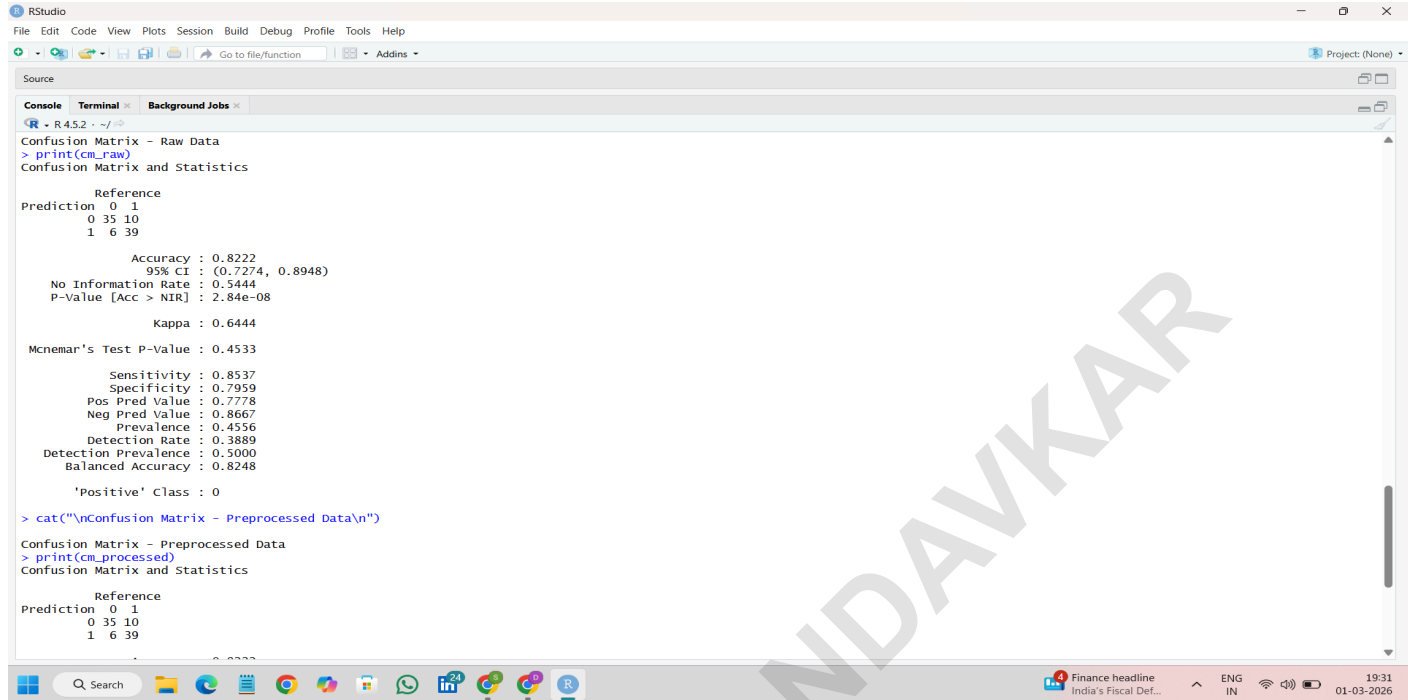
Confusion Matrix - Raw Data
> print(cm_raw)
```

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RStudio interface showing the console output for a confusion matrix analysis. The console displays the following text:

```
Confusion Matrix - Raw Data
> print(cm_raw)
Confusion Matrix and Statistics

      Reference
Prediction 0 1
          0 35 10
          1  6 39

      Accuracy : 0.8222
      95% CI : (0.7274, 0.8948)
      No Information Rate : 0.5444
      P-Value [Acc > NIR] : 2.84e-08

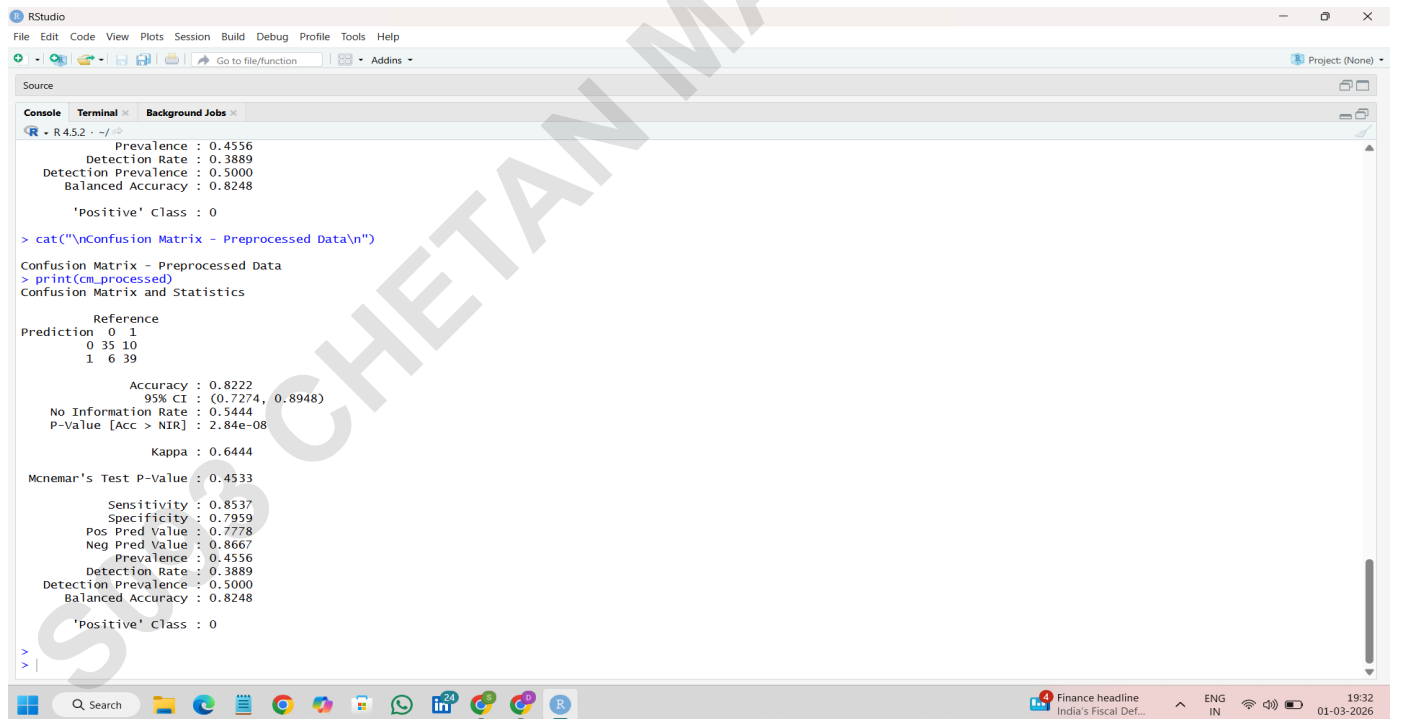
      Kappa : 0.6444

      McNemar's Test P-Value : 0.4533
      Sensitivity : 0.8537
      Specificity : 0.7959
      Pos Pred Value : 0.7778
      Neg Pred Value : 0.8667
      Prevalence : 0.4556
      Detection Rate : 0.3889
      Detection Prevalence : 0.5000
      Balanced Accuracy : 0.8248

      'Positive' Class : 0

> cat("\nConfusion Matrix - Preprocessed Data\n")
Confusion Matrix - Preprocessed Data
> print(cm_preprocessed)
Confusion Matrix and Statistics

      Reference
Prediction 0 1
          0 35 10
          1  6 39
```



RStudio interface showing the console output for a confusion matrix analysis. The console displays the following text:

```
Prevalence : 0.4556
Detection Rate : 0.3889
Detection Prevalence : 0.5000
Balanced Accuracy : 0.8248

'Positive' Class : 0

> cat("\nConfusion Matrix - Preprocessed Data\n")
Confusion Matrix - Preprocessed Data
> print(cm_preprocessed)
Confusion Matrix and Statistics

      Reference
Prediction 0 1
          0 35 10
          1  6 39

      Accuracy : 0.8222
      95% CI : (0.7274, 0.8948)
      No Information Rate : 0.5444
      P-Value [Acc > NIR] : 2.84e-08

      Kappa : 0.6444

      McNemar's Test P-Value : 0.4533
      Sensitivity : 0.8537
      Specificity : 0.7959
      Pos Pred Value : 0.7778
      Neg Pred Value : 0.8667
      Prevalence : 0.4556
      Detection Rate : 0.3889
      Detection Prevalence : 0.5000
      Balanced Accuracy : 0.8248

      'Positive' Class : 0

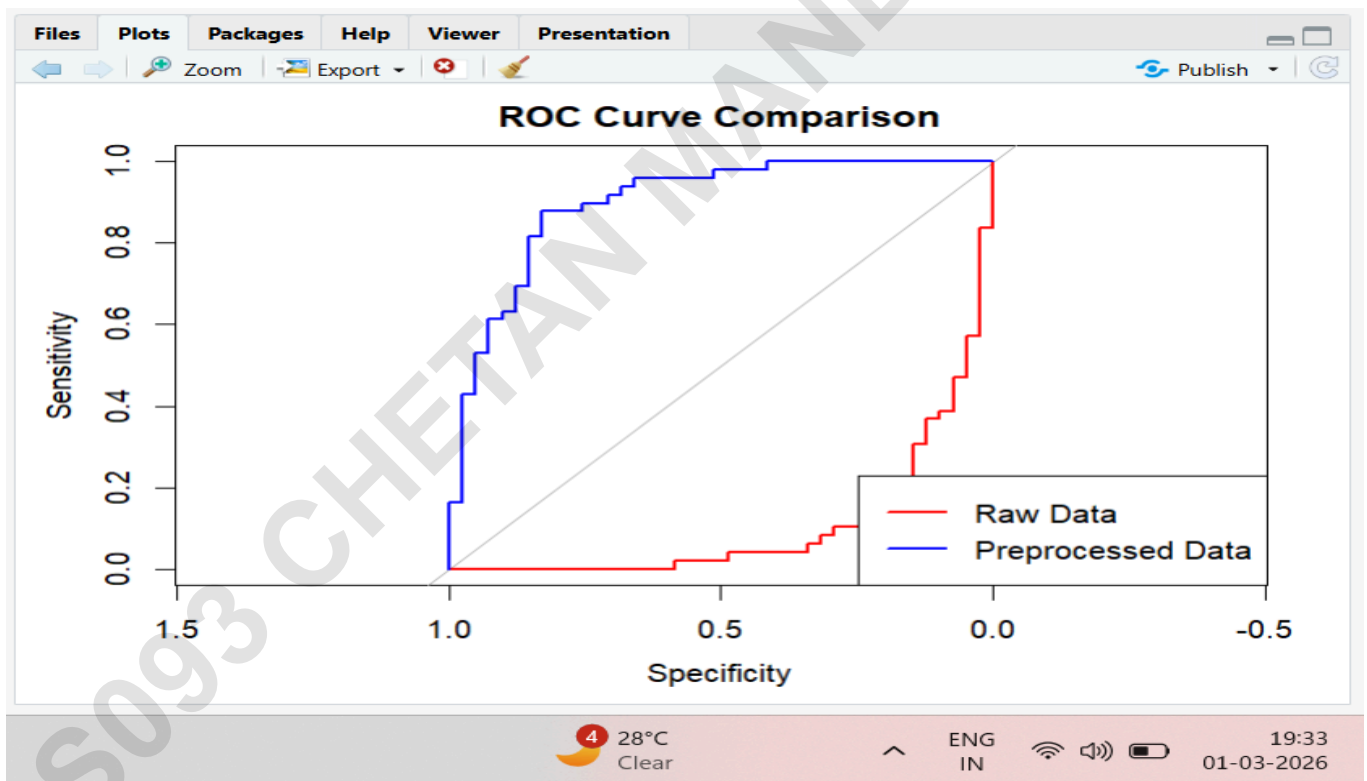
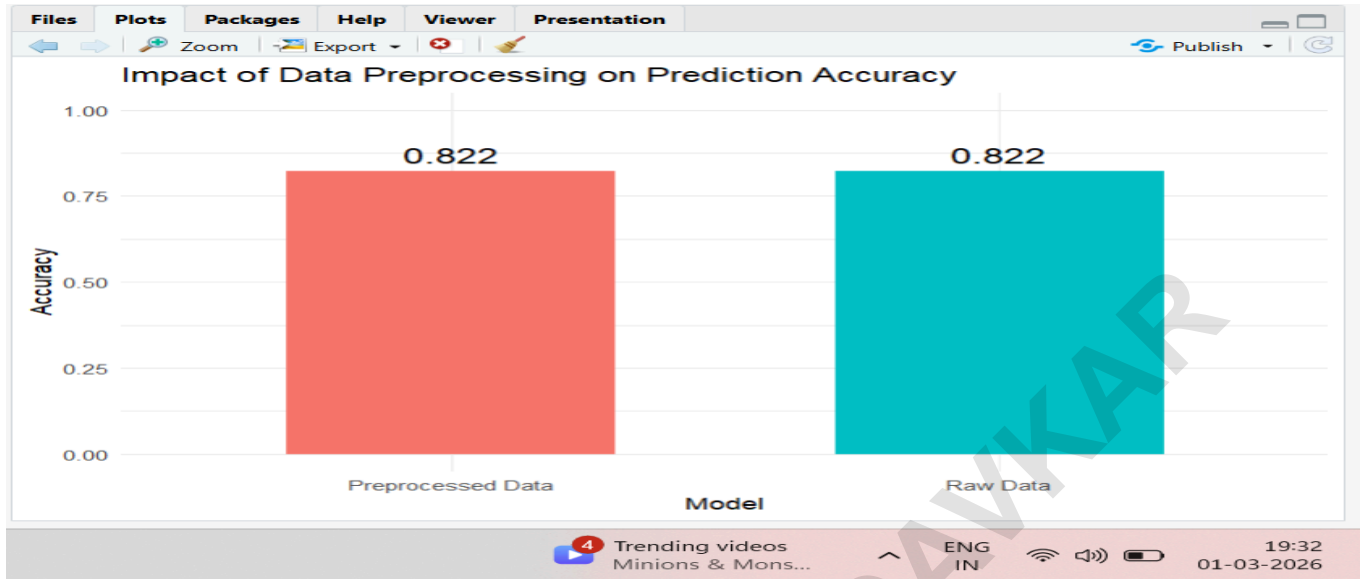
>
> |
```

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